Do Now · retrieval

Lesson 128 · Run-Length Encoding: Encoding and Decoding

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

When must lossless compression be used?

Q2

In a queue, what does FIFO stand for?

Q3

What is a function parameter?

Q4

In a BST, which side holds a smaller value?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Identify runs — Group identical adjacent symbols, not identical symbols separated by others.

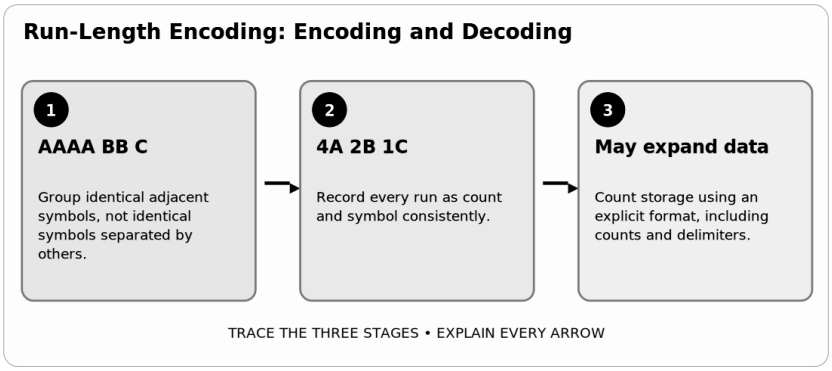
Core idea 2

Encode — Record every run as count and symbol consistently.

Core idea 3

Check cost — Count storage using an explicit format, including counts and delimiters.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

RLE — Encodes consecutive equal symbols as a symbol and count.

run — A consecutive sequence of identical symbols.

overhead — Extra data required by an encoding scheme.

lossless — Decoding reproduces the original exactly.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Encode AAAAABBCCC using (count,symbol) pairs; verify the result.

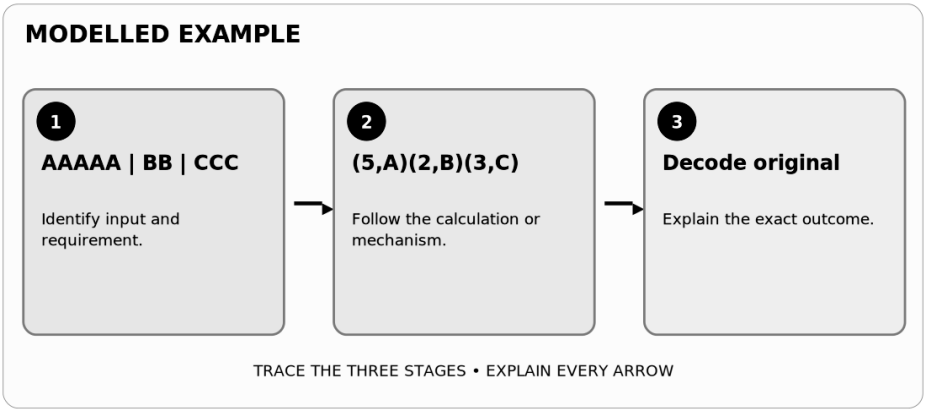
Follow the worked method

1. AAAAA | BB | CCC — Predict the first step from the prompt.

2. (5,A)(2,B)(3,C) — Explain the intermediate result or decision.

3. Decode original — AAAAA =5A, BB=2B, CCC=3C. Encoded result (5,A)(2,B)(3,C); decoded data restores AAAAABBCCC exactly.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

A B A B A B

Predict the first step from the prompt.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Apply RLE to AAAABBBCCAAAA using count-symbol pairs. [6]

Question 2

Decode (3,X)(1,Y)(2,X) and explain why runs must retain order. [5]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A bitmap row has 40 alternating pixels. Explain why RLE might increase file size and identify a better-suited pattern. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

RLE always reduces the size of any input.

Use key vocabulary from memory

1. Explain RLE in your own words.

2. Explain run in your own words.

3. Explain overhead in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.